

# Recommended Assessment

## Push Button

### Questions

1. What happens to each signal when each button is pressed or released? What happens when each button is held down?
2. After setting the polarity of Button 0 to 0 in the initialization, what happens to each signal when each button is pressed or released? What happens when each button is held down?
3. If we want to use Button 1 as a toggle switch for **buttonLatch**, what should both signals look like on the scope?
4. What does the scope show when **buttonLatch** is toggled if the value of Button 1 is 1? Attach your screenshot to show your results.
5. Based on the terminal output, what happens to buttonLatch when you press Button 1? What needs to change if we only want to toggle buttonLatch once each time Button 1 is pressed?
6. After making changes to the code implementation so that buttonLatch successfully toggles once each time Button 1 is pressed, how did you verify the implementation using the terminal output and scope? Take a screenshot of the scope to show that the value of buttonLatch successfully toggles each time Button 1 is pressed.
7. Suppose the polarity of Button 1 is reversed and we want to use Button 1 as a toggle switch for buttonLatch. How would the implementation change?
8. When the sampling time of the probe is changed to every 10 samples, is the terminal output consistent with the scope output? How are they different? Why?
9. What do you notice after you changed the sampling frequency to 400 Hz?

## Push Button Cheat Sheet

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|----------------------------------------|--|
| What is it?                            |  |
| What does it measure directly?         |  |
| What can you measure indirectly?       |  |
| Where are they used?                   |  |
| Pros                                   |  |
| Cons                                   |  |
| Notes/<br>Important things to remember |  |